

Software Test Plan (STP)

Bank Management System

Project: Bank Management System

Version: 1.0

Date: 15-09-2026

Status: Draft

Team Members

Kruthik Reddy S: PES2UG24CS241

Kiran B G: PES2UG24CS227

Kotapalli Devi Sai Charan: PES2UG24CS231

1. Introduction

Purpose: This document defines the test plan for the Bank Management System v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers core banking operations including customer/account authentication, cash deposit, cash withdrawal, and balance inquiry. External core-banking infrastructure, physical hardware, and third-party payment gateway internals are excluded.

References: Bank Management System SRS v1.0, Design Specifications v1.0, Banking Standards, PCI-DSS.

Definitions: SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), PIN (Personal Identification Number).

2. Test Items

- Customer Authentication module (login / account verification)
- Deposit module
- Withdrawal module
- Balance Inquiry module
- Account Management interface

3. Features to be Tested

Features mapped to SRS requirement IDs:

Requirement ID	Feature
BMS-F-001	Validate customer login credentials
BMS-F-010	Deposit funds into an account
BMS-F-020	Withdraw funds with balance check
BMS-F-030	Balance inquiry for an account
BMS-NF-001	Response time ≤ 5s
BMS-NF-002	99.9% system availability

4. Features Not to be Tested

- Core banking backend / database engine internals (assumed pre-tested)
- Third-party payment gateway integrations
- Physical hardware (if applicable) and network infrastructure

5. Test Approach / Strategy

Levels

- Unit tests (module-level)
- Integration tests (module-to-module and module-to-database)
- System tests (end-to-end banking workflows)
- Acceptance tests (UAT)

Types

- Functional testing (core features: deposit, withdrawal, balance inquiry)
- Regression testing
- Performance testing (response time, load)
- Usability testing (UI clarity, accessibility)

Entry Criteria: Stable build delivered, test data available, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

5.1 Security Validation

- Validate credential handling (masking, no logging of sensitive data)
- TLS 1.2+ verification for data in transit
- PCI-DSS compliance checks where applicable
- Input validation / fuzzing on account and transaction fields
- Basic penetration checks on the authentication flow

6. Test Environment

Hardware: Standard development/test workstation; no specialized banking hardware required for this mini project.

Software: Bank Management System application v1.0, backend database (test instance).

Tools: Manual test execution, IDE debugger; optionally Selenium (UI automation), Postman (API), JMeter (performance).

Test Data: Dummy customer accounts and sample transactions.

7. Test Schedule

Milestone	Target Date
Test case design	20-Sep-2026
Environment setup	22-Sep-2026
Test execution start	23-Sep-2026
Test execution end	30-Sep-2026
UAT	01-Oct-2026 to 03-Oct-2026

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Responsibilities are distributed across the three-member team so that each member owns end-to-end testing (design, execution, and defect logging) for specific modules, in addition to shared plan-level duties.

Name	SRN	Primary Role	Responsibility
Kruthik Reddy S	PES2UG24CS241	QA Lead & Test Engineer – Authentication	Prepare and maintain the test plan; coordinate overall test execution and schedule. Design and execute test cases for the Authentication module; log related defects.
Kiran B G	PES2UG24CS227	Test Engineer – Deposit & Withdrawal	Design and execute test cases for the Deposit and Withdrawal modules. Log defects for these modules and verify fixes after resolution.
Kotapalli Devi Sai Charan	PES2UG24CS231	Test Engineer – Balance Inquiry & Support	Design and execute test cases for the Balance Inquiry module. Support developers during defect triage and assist in regression testing across modules.

Shared Responsibilities

- All three members jointly review and approve the final Test Summary Report before sign-off.
- Each member acts as developer for their own module and as an independent reviewer for one peer's module, to keep testing objective.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds from the development team.

Risk	Mitigation
Limited team size (3 members) causing schedule overlap	Distribute modules early and track progress with a shared checklist.
Test environment downtime	Maintain a backup local/cloud environment.
Dependency on shared database/test data setup	Prepare and version-control dummy test data in advance.

11. Assumptions & Dependencies

- The application backend/database will be stable and available during test execution.
- Test data (dummy accounts/transactions) will be prepared before execution begins.
- All three team members will be available through the scheduled test window.

12. Suspension & Resumption Criteria

Suspend testing if:

- Environment is unavailable for more than 4 hours.
- Build is too unstable (blocks more than 30% of planned test cases).

Resume testing if:

- Blocking defects are resolved.
- Environment is stabilized.

13. Test Case Management & Traceability

The Requirements Traceability Matrix (RTM) maps SRS requirements to test cases. Example:

Requirement ID	Feature	Test Case ID(s)	Owner
BMS-F-001	Login validation	TC-Auth-01, TC-Auth-02	Kruthik Reddy S
BMS-F-010	Deposit	TC-Dep-01, TC-Dep-02	Kiran B G
BMS-F-020	Withdrawal	TC-WD-01, TC-WD-02	Kiran B G
BMS-F-030	Balance inquiry	TC-Bal-01	Kotapalli Devi Sai Charan
BMS-NF-001	Response time	TC-Perf-01	Kotapalli Devi Sai Charan

14. Test Metrics & Reporting

Metrics collected

- % test cases executed
- % passed / failed

- Defect density
- Defect aging
- Requirement coverage

Reports

- Periodic execution status update (shared among team members)
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Kruthik Reddy S	
Test Engineer	Kiran B G	
Test Engineer	Kotapalli Devi Sai Charan	